

Progress Report 3 - Fourier Transform

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Since the previous progress report, the main DFT processing and output features of the program have been implemented and tested. The program can now successfully extract the samples from x.signal and calculate the DFT over the frequency range specified through the program arguments. The Basic Test using x.signal 32 4 8 8 successfully extracted all 32 samples and produced the expected DFT results.

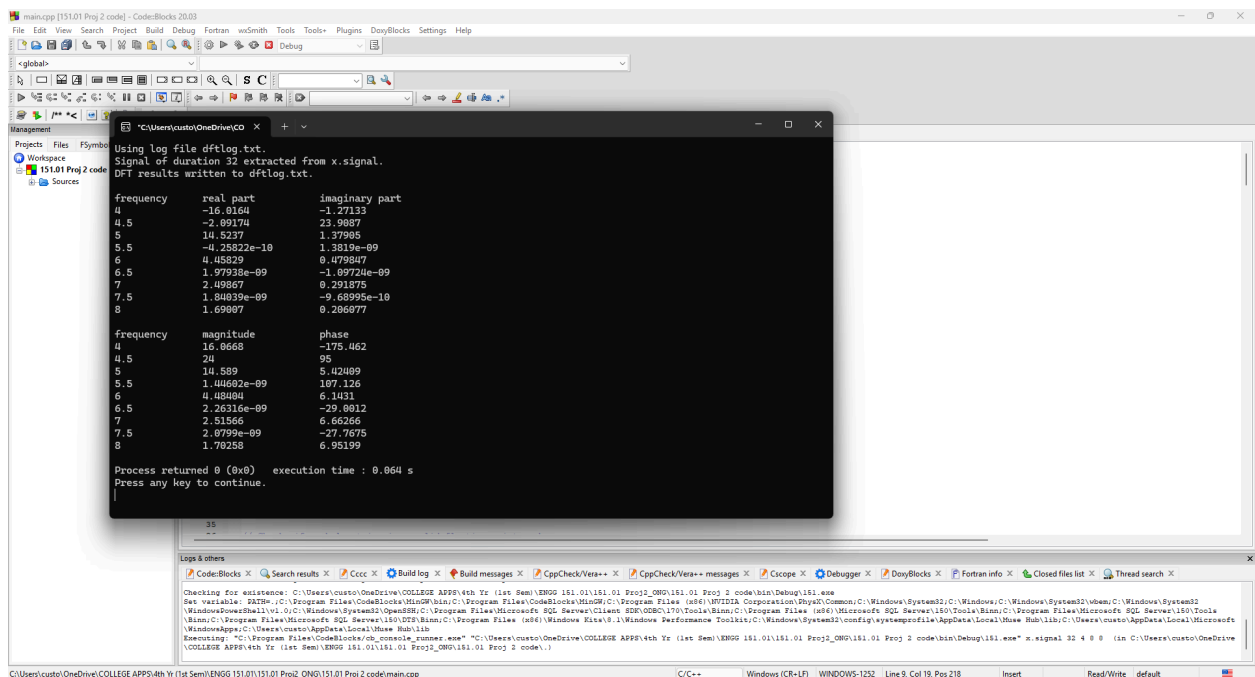


Figure 1. Basic Test Results

The output was also improved to separately display the real and imaginary components and the magnitude and phase for each frequency. Results are written to dtflog.txt by default, with support for a user-specified logfile and appending results to an existing file.

File	Edit	View
frequency	Real	Imaginary
4	-16.0164	-1.27133
4.5	-2.09174	23.9087 24
5	14.5237 1.37905	14.589 5.42409
5.5	-4.25822e-10	1.3819e-09
6	4.45829 0.479847	4.48404 6.1431
6.5	1.97938e-09	-1.09724e-09
7	2.49867 0.291875	2.51566 6.66266
7.5	1.84839e-09	-9.68995e-10
8	1.69007 0.206077	1.70258 6.95199
frequency	real part	imaginary part
4	-16.0164	-1.27133
4.5	-2.09174	23.9087
5	14.5237 1.37905	
5.5	-4.25822e-10	1.3819e-09
6	4.45829 0.479847	
6.5	1.97938e-09	-1.09724e-09
7	2.49867 0.291875	
7.5	1.84839e-09	-9.68995e-10
8	1.69007 0.206077	
frequency	magnitude	phase
4	16.0668	-175.462
4.5	24	95
5	14.589 5.42409	
5.5	1.44602e-09	107.126
6	4.48404	6.1431
6.5	2.26316e-09	-29.0012
7	2.51566	6.66266
7.5	2.0799e-09	-27.7675
8	1.70258	6.95199
frequency	real part	imaginary part
4	-16.0164	-1.27133
4.5	-2.09174	23.9087
5	14.5237	1.37905
5.5	-4.25822e-10	1.3819e-09
6	4.45829	0.479847
6.5	1.97938e-09	-1.09724e-09
7	2.49867	0.291875
7.5	1.84839e-09	-9.68995e-10
8	1.69007	0.206077
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4	16.0668	-175.462
4.5	24	95
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5.5	1.44602e-09	107.126
6	4.48404	6.1431
6.5	2.26316e-09	-29.0012
7	2.51566	6.66266
7.5	2.0799e-09	-27.7675
8	1.70258	6.95199
frequency	real part	imaginary part
4	-16.0164	-1.27133
4.4	-9.27989	21.9573
4.8	15.0563	13.2083
5.2	7.14473	-4.19475
5.6	0.409176	1.90501
6	4.45829	0.479847
6.4	0.43882	-0.968423
6.8	1.64926	1.51531
7.2	1.62468	-0.910242
7.6	0.117027	0.594606
8	1.69007	0.206077
frequency	magnitude	phase
4	16.0668	-175.462
4.4	23.0378	112.911
4.8	20.0207	41.2592
5.2	8.28511	-30.4177
5.6	1.94845	77.8776
6	4.48404	6.1431
6.4	1.06321	-65.6234
6.8	2.23969	42.5762
7.2	1.86229	-29.2661
7.6	0.606013	78.8657
8	1.70258	6.95199
frequency	real part	imaginary part
4	-16.0164	-1.27133
4.4	-9.27989	21.9573
4.8	15.0563	13.2083
5.2	7.14473	-4.19475
5.6	0.409176	1.90501
6	4.45829	0.479847
6.4	0.43882	-0.968423
6.8	1.64926	1.51531
7.2	1.62468	-0.910242
7.6	0.117027	0.594606

Figure 2. DFT Log File

Additional testing was performed to check input handling. Invalid command-line values were successfully identified, and testing with an invalid value inside a signal file showed that the program stops extraction at the invalid entry while retaining the valid samples before it.

In this stage, the main program meets the project requirements. The remaining work will focus on further testing the code.